

ACTG - Sensitivity analysis

We have

$$\begin{aligned}
p(m | D) &= \frac{Z_m(D)p(m)}{\sum_{j \in \mathcal{M}} Z_j(D)p(j)}, \\
&= \frac{Z_m(D) \frac{Z_j(D_0)}{\sum_{k \in \mathcal{M}} Z_k(D_0)}}{\sum_{j \in \mathcal{M}} Z_j(D) \frac{Z_m(D_0)}{\sum_{k \in \mathcal{M}} Z_k(D_0)}}, \\
&= \frac{Z_m(D)Z_m(D_0)}{\sum_{j \in \mathcal{M}} Z_j(D)Z_j(D_0)}, \\
&= \frac{\int L(\beta^{(m)}|D)\pi(\beta^{(m)}, a_0 | D_0) d\beta^{(m)} da_0 \int L(\beta^{(m)}|D_0)\pi_0(\beta^{(m)} | \psi_0) d\beta^{(m)}}{\sum_{j \in \mathcal{M}} \int L(\beta^{(j)}|D)\pi(\beta^{(j)}, a_0 | D_0) d\beta^{(j)} da_0 \int L(\beta^{(j)}|D_0)\pi_0(\beta^{(j)} | \psi_0) d\beta^{(j)}}, \\
&= \frac{\int L(\beta^{(m)}|D) \frac{L(\beta^{(m)}|D_0)^{a_0}}{C(a_0)} \pi_A(a_0) \pi_0(\beta^{(m)} | \tau_0) d\beta^{(m)} da_0 \int L(\beta^{(m)}|D_0)\pi_0(\beta^{(m)} | \psi_0) d\beta^{(m)}}{\sum_{j \in \mathcal{M}} \int L(\beta^{(j)}|D) \frac{L(\beta^{(j)}|D_0)^{a_0}}{C(a_0)} \pi_A(a_0) \pi_0(\beta^{(j)} | \tau_0) d\beta^{(j)} da_0 \int L(\beta^{(j)}|D_0)\pi_0(\beta^{(j)} | \psi_0) d\beta^{(j)}},
\end{aligned}$$

where $a_0 \sim \text{Beta}(\delta_0, \lambda_0)$ and $\beta^{(m)} | \phi \sim N(\mathbf{0}, \phi^2 \cdot \mathbf{I}_{p^{(m)}})$.

We start by computing probabilities for several values of ψ_0 , $(\delta_0, \lambda_0) = (15, 15)$ and $\tau_0 = \sqrt{5}$:

	model	$p(m)$	$\log(p(D m))$	$p(m D)$
1	age, treatment, cd4	0.49	-34.92	0.52
2	age, treatment, cd4	0.49	-34.86	0.53
3	age, treatment, cd4	0.49	-34.81	0.55

Then, we compute probabilities for several values of τ_0 , $(\delta_0, \lambda_0) = (15, 15)$ and $\psi_0 = \sqrt{5}$:

Lastly, we compute probabilities for $(\delta_0, \lambda_0) \in \{(1, 1), (5, 5), (15, 15), (50, 1)\}$, $\psi_0 = \sqrt{5}$ and $\tau_0 = \sqrt{10}$:

Best Model	$p(m)$	$\log(Z_m(D))$	$p(m D)$	(δ_0, λ_0)
age, treatment, cd4	0.27	-34.87	0.30	(1, 1)
age, treatment, cd4	0.27	-34.81	0.31	(5, 5)
age, treatment, cd4	0.27	-34.81	0.31	(15, 15)
age, treatment, cd4	0.27	-34.87	0.33	(50, 1)

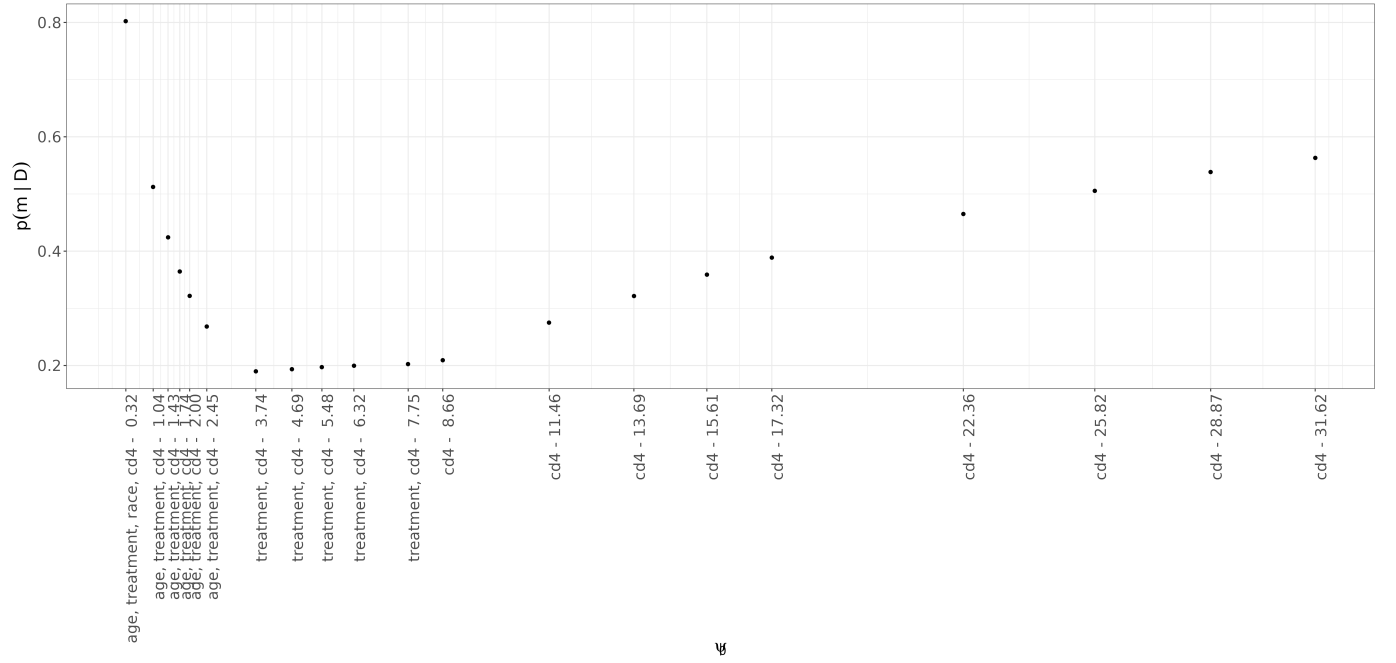


Figure 1: Posterior probability of the best model for several values of ψ_0 .

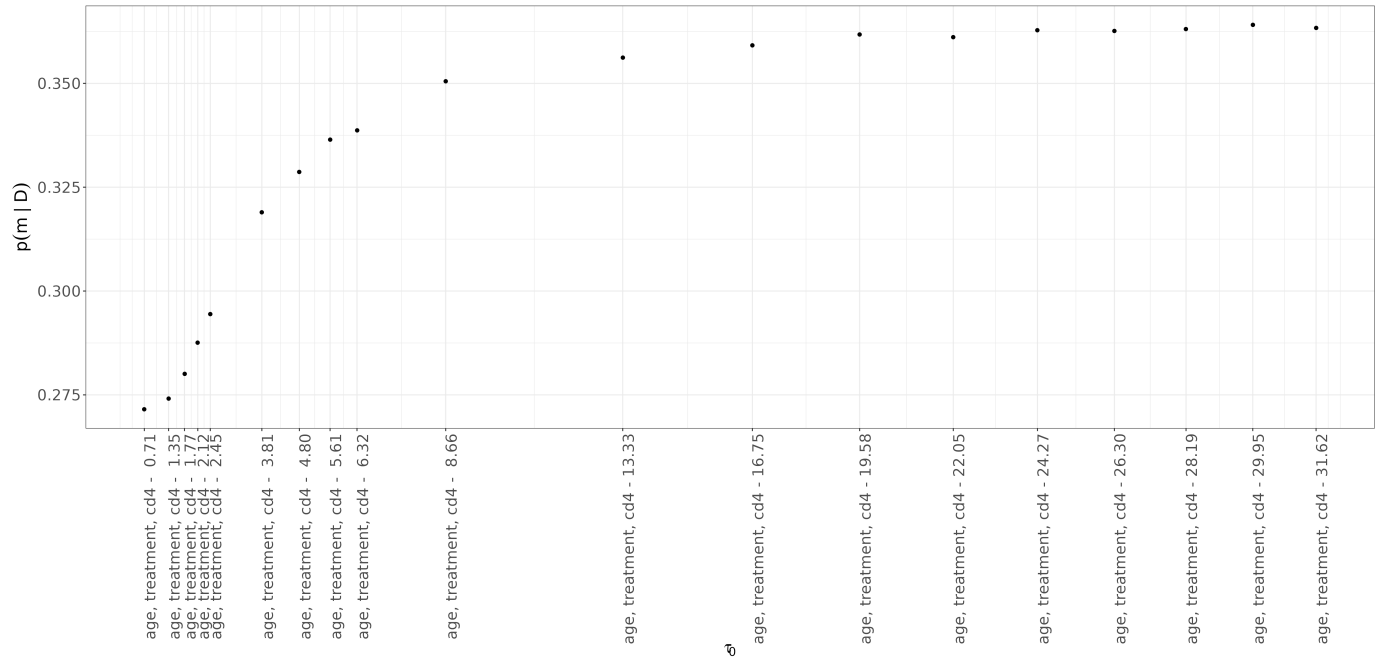


Figure 2: Posterior probability of the best model for several values of τ_0 .